

Day 5: Troubleshooting Network Services Practical (1 hour): Troubleshoot DNS and DHCP issues in a simulated network. Task: Document solutions for common DNS and DHCP issues.

Task 1.

Simulate a DNS resolution failure on your local machine by editing the hosts file to point `www.spotify.com` to a wrong IP address, then try to ping the domain and document the error message you receive.

1. Open Notepad as Administrator.
2. Navigate to `C:\Windows\System32\drivers\etc\hosts`.
3. Add entry:
Code: `123.123.123.123 www.spotify.com`
4. Save and close the file.
5. Open Command Prompt.
6. Type: `ping www.spotify.com`.
7. Observe the output → it resolves to wrong IP and shows “Request timed out”

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

```
Pinging atc.spotify.map.fastly.net [2a04:4e42:31::810] with 32 bytes of data:
Reply from 2a04:4e42:31::810: time=59ms
Reply from 2a04:4e42:31::810: time=79ms
Reply from 2a04:4e42:31::810: time=38ms
Reply from 2a04:4e42:31::810: time=49ms

Ping statistics for 2a04:4e42:31::810:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 38ms, Maximum = 79ms, Average = 56ms
```

Task 2.

Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

Step 1: Configure DHCP on Router

```
enable
configure terminal
```

Create DHCP pool with wrong gateway:

```
ip dhcp pool STUDENT
network 192.168.10.0 255.255.255.0
default-router 192.168.20.1 ← (Wrong Gateway)
dns-server 8.8.8.8
```

Assign IP to router interface

```
interface gigabitEthernet 0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
```

Step 2: Connect Client PC

- Go to **PC → Desktop → IP Configuration**
- Select **DHCP**
- The PC will receive:
IP: 192.168.10.x
Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.20.1 (wrong)

For Verification

```
ipconfig /all
```

Task 3.

Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

Step 1: Verify Connectivity

On client PC

ping 8.8.8.8

Works : Internet connectivity is fine.

Fails for domain names : DNS issue suspected.

Step 2: Use nslookup

nslookup www.google.com

- If output shows “DNS request timed out” or “Server not found”, DNS server is misconfigured.
- Check which DNS server is being used (shown in nslookup output).

Step 3: Use dig (if available)

dig www.google.com

- Look at ANSWER SECTION
- If empty → DNS server not responding.
- If response comes from wrong server → misconfiguration

Step 4: Check Client DNS Settings

On PC → Desktop → IP Configuration:

- Verify DNS server address.
- Check which DNS server is being used (shown in nslookup output).
- If blank or incorrect, DNS queries fail.

Step 5: Check DHCP Server Configuration

show running-config

- Look at DHCP pool settings.
- Example of wrong config:

dns-server 192.168.20.1 (Incorrect)

dns-server 8.8.8.8

Solution

1. Correct DHCP pool configuration

Configure terminal

Ip dhcp pool student

dns server 8.8.8.8

2. Release & renew IP

Ipconfig / release

Ipconfig / renew

dns server 8.8.8.8

3. Test again

Nslookup www.google.com

ping www.google.com

Task 4.

A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it.

Hint: Check for network connectivity, DHCP scope exhaustion, and server availability.

A device is showing an APIPA address (169.254.x.x) instead of receiving an IP address from the DHCP server

An APIPA (Automatic Private IP Addressing) address in the range 169.254.0.1 – 169.254.255.254 indicates that the device tried to obtain an IP address from the DHCP server but did not receive one.

Possible Causes

1. Network cable disconnected or wireless connection not established.
2. DHCP server is turned off or unavailable.
3. DHCP service is stopped on the server.
4. DHCP scope is exhausted, so no IP addresses are available.
5. Incorrect VLAN or switch port configuration preventing communication with the DHCP server.
6. Router or DHCP relay is not configured (if the server is on another network).
7. Firewall or ACL blocking DHCP traffic (UDP ports 67 and 68).
8. Client network adapter is disabled or has an incorrect configuration.
9. Incorrect subnet mask or gateway configured on the DHCP server.
10. IP address conflict or DHCP database issues.

Troubleshooting Process

Step 1: Verify Physical Connectivity

- Check that the Ethernet cable is connected properly.
- Ensure switch and router interfaces are up (green in Cisco Packet Tracer).
- If using Wi-Fi, verify the device is connected to the correct network.

Step 2: Check the Client IP Address

On the client PC, open Command Prompt and type:

```
ipconfig /all
```

If the IP address is 169.254.x.x, the client has not received an IP address from the DHCP server.

Step 3: Renew the IP Address

Run:

ipconfig /release

ipconfig /renew

If the client still receives an APIPA address, continue troubleshooting.

Step 4: Verify DHCP Server Status

- Ensure the DHCP server is powered on.
- Check that the DHCP service is running.
- In Cisco Packet Tracer:
 - Open the DHCP Server.
 - Go to Services > DHCP.
 - Verify the DHCP service is On.

Step 5: Check DHCP Scope

Verify:

- Network address is correct.
- Subnet mask is correct.
- Default gateway is correct.
- DNS server is configured (optional).
- Available IP addresses remain in the pool.

If the scope is full, increase the address range or remove unused leases.

Step 6: Test Network Connectivity

From the client, ping the default gateway:

ping 192.168.1.1

If the ping fails:

- Check cables.
- Verify switch ports.
- Ensure router interfaces are enabled.

Step 7: Verify Router Configuration

If the DHCP server is on another network, configure a DHCP relay:

```
Router(config)# interface g0/0
```

```
Router(config-if)# ip helper-address <DHCP_Server_IP>
```

Step 8: Check Firewall or ACL

Ensure DHCP traffic is not blocked.

Required ports:

- UDP 67 (DHCP Server)
- UDP 68 (DHCP Client)

Step 9: Restart the Client

Restart the PC or disable and re-enable the network adapter, then renew the IP address.

Step 10: Verify the Result

Run:

```
ipconfig
```

The client should now receive a valid IP address, for example:

IPv4 Address : 192.168.1.10

Subnet Mask : 255.255.255.0

Default Gateway : 192.168.1.1

Task 5.

Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

Common DNS and DHCP Issues

Protocol	Issue	Symptoms	Possible Causes	Recommended Solution
DNS	DNS server not responding	Websites cannot be opened using domain names, but IP addresses work	DNS server is down, incorrect DNS server address, network connectivity issue	Verify the DNS server is running, check DNS settings, and test connectivity using ping or nslookup.
DNS	Incorrect DNS records	Users are redirected to the wrong website or receive "Host not found" errors	Wrong A, AAAA, or CNAME records, outdated DNS entries	Update the DNS records, clear the DNS cache (ipconfig /flushdns), and verify the zone configuration.
DNS	Slow DNS resolution	Websites take a long time to load	High DNS server load, network latency, or incorrect forwarder configuration	Use a reliable DNS server, optimize DNS settings, and verify network connectivity.
DHCP	Client receives APIPA address (169.254.x.x)	Device cannot communicate on the network and has no valid IP address	DHCP server unavailable, cable disconnected, DHCP relay missing, or scope exhausted	Check network connectivity, verify the DHCP server is running, renew the IP address, and ensure addresses are available in the DHCP scope.

Protocol	Issue	Symptoms	Possible Causes	Recommended Solution
DHCP	DHCP scope exhausted	New devices fail to obtain an IP address	All available IP addresses in the DHCP pool have been assigned	Increase the DHCP scope, reduce lease duration, or remove inactive leases.
DHCP	Incorrect DHCP options	Client receives an IP address but cannot access other networks or the Internet	Wrong default gateway, subnet mask, or DNS server configured in DHCP	Correct the DHCP scope options, renew the client IP address, and verify network connectivity.